Filing Impact Evaluator - Requirements

Version 1.27 | 29 Oct 2026 | Companion file: DECISIONS.md (decision log). Research project, not financial advice.

1. Goal

Measure whether an LLM agent can predict the direction and size of a stock's move after an NSE/BSE corporate announcement, and whether it beats simple baselines. The system does **not** place orders. It produces timestamped signals and a trustworthy scorecard. A trader app can be added later by reading the stored signals.

2. Scope

• **In:** announcement polling for hundreds of NSE/BSE stocks, LLM signals, pre-announcement move features, Telegram alerts, outcome evaluation, scorecard, read-only signals API.
• **Out (v1):** order placement, UI dashboard, portfolio management, options/derivatives, any use of non-public information.

3. Features

ID	Feature	Description	Priority
F1	Announcement poller (Go)	Poll BSE + NSE exchange-wide feeds every few seconds; dedupe; event time = earliest of the two; store exchange and first-seen times.	Must
F2	Storage	SQLite first (Postgres later). Raw filings, signals, prices, outcomes. Source of truth.	Must
F3	Pre-filter classifier	Cheap LLM tags category and drops routine filings (board meeting notices etc).	Must
F4	LLM analyzer	Reads interesting filings; returns strict JSON signal: direction, confidence, expected horizon, reason. Provider interface: mock, Ollama, Claude.	Must
F5	Telegram notifier	Msg 1: prediction on arrival. Msg 2: verdict as reply after outcome. Daily digest. Filtered by category/confidence.	Must
F6	Evaluator	Joins signals to actual returns at 15m, 1h, EOD, 1d, 5d. Strict no look-ahead.	Must
F7	Scorecard + baselines	Hit rate, avg return, calibration per category and timing bucket; compared with always-up, keyword rule, no-trade.	Must
F8	Pre-move feature engine	Features computed from public price/volume before the announcement (see section 5).	Should
F9	Combiner model	Logistic regression / gradient boosting: LLM signal + pre-move features -> is move already priced in?	Should
F10	Signals API	Read-only table/endpoint with versioned schema for the future trader app.	Should
F11	Historical backfill	Pipeline test and rough sanity check only; results labeled "possibly contaminated".	Should
F12	Ops	VPS deployment, restart on failure, gap detection, logs, latency metrics.	Should

4. Data flow

Poller -> store -> pre-filter -> analyzer -> signal saved -> Telegram (prediction) -> evaluator at each horizon -> outcome saved -> Telegram (verdict) -> scorecard / digest. Telegram is only a view; every result lives in the database.

Timestamp	Meaning	Why it matters
exchange_time	When the exchange published it	Defines the true start of the price reaction
first_seen_time	When our poller detected it	Measures our latency vs the exchange
signal_time	When the signal was created	Measures total delay; needed for realistic trading later

5. Pre-announcement abnormal-move features (public data only)

• Return over previous 1, 3, 5 days minus market and sector return.
• Volume and delivery-percentage spike versus the 20-day average.
• Price move in the hour before the announcement.
• Whether the date was pre-scheduled (e.g. known results date) vs surprise.
• Timing bucket: market hours, pre-open, after close.
• Question answered: if the stock already moved in the signal direction, is the remaining move smaller or does it continue? Cause of any move is unknown; this is not an insider detector.

6. Expected outcomes and success metrics

Metric	Meaning / target
Ingest latency	exchange_time to first_seen_time: report median and p95. No fixed target until measured.
Coverage	% of announcements captured vs the exchange list; gaps are logged.
Hit rate vs baseline	Direction accuracy per category and timing bucket, compared with baselines.
Calibration	Do higher-confidence signals have higher accuracy?
Net return	Average return after estimated costs (brokerage, taxes, slippage), not gross.
Priced-in test	Does the pre-move feature set improve predictions over the LLM alone?
Sample size (proposal)	At least 300 live signals per bucket before drawing conclusions; use a statistical test, not eyeballing.
Valid outcomes	Positive edge, no edge, or negative edge are all acceptable results if measured honestly.

7. Future trader-app requirements (design now, build later)

• Reads only the signals store/API; no trading code inside the core system.
• Signal schema is versioned and includes: symbol, filing id, three timestamps, direction, confidence, horizon, category, pre-move features, model name/version, historical accuracy of that bucket.
• Must start with paper trading; live orders only after live results justify it.
• Needs risk controls: position size limit, daily loss limit, liquidity filter, kill switch, idempotent order ids.
• Evaluation must already include costs and liquidity so backtest numbers are realistic.

8. Risks and mitigations

Risk	Mitigation
LLM memorized past outcomes (contamination)	Trust live data only; label history as contaminated.
Unofficial exchange endpoints break or block polling	Cache, back off, monitor gaps; verify early with a spike script.
Historical timestamps may not be true publish time	Verify first; use daily returns as the main metric if unreliable.
First move happens before we can act	Evaluate timing buckets separately; focus on after-hours announcements.
Laptop downtime creates data gaps	Run on a VPS.
LLM cost grows with volume	Two-tier models; measure one real day first.

9. Milestones

#	Milestone	Done when
M1	Spike (done)	5 stocks fetched; fields and timestamps understood
M2	Live collector (done, validated 1 day)	All stocks via exchange-wide feeds polled; stored with timestamps; latency logged
M3	Signals (started)	Filter and analyzer (mock, Gemini) built; Telegram predictions next
M4	Evaluation	Verdict replies, scorecard, baselines
M5	Claude run	Real evaluation with Claude on live data
M6	Pre-move model + API	Features, combiner, read-only signals API

10. Keeping documents current (maintenance rule)

• **DECISIONS.md is the single source of truth** and is self-contained: it holds these requirements, decisions with reasons, changelog, status, next steps, unverified assumptions and instructions for any LLM. This PDF is a readable copy.
• At the start of every new session (any LLM), paste DECISIONS.md in full. The LLM states status and next step, then continues.
• During the session the LLM adds each decision and change to the file immediately, with the reason. Old decisions are never deleted; they are marked superseded by a new one.
• At the end of the session (or when asked to wrap up, or when tokens may run out) the LLM outputs the complete updated file, bumps the version, updates status and next steps, and refreshes this PDF if it can create files.
• Priority rule: if the token budget may be nearly used up, the LLM stops the current task and outputs the complete updated DECISIONS.md (and PDF if possible) first, even if work is unfinished.
• Assumptions stay listed as unverified until actually tested.